

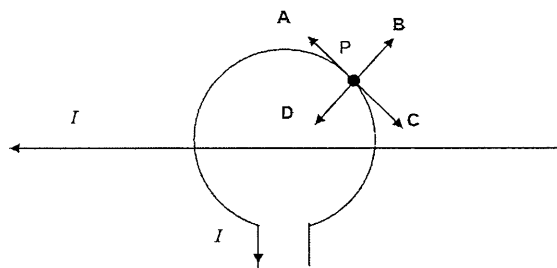
Important note: *denotes questions that may require knowledge from later chapters in order to solve them.

I. MCQs ($\mu_0 = 4\pi \times 10^{-7} \text{ H m}^{-1}$)

1. [AJC/H2/Prelims 2010/P1/29]

A long horizontal straight wire is placed close and parallel to the plane of a coil as shown in the diagram. Both the coil and the wire carry current I in the directions indicated.

What is the direction of the force at point P on the coil due to these currents?



2. [AJC/H2/Prelims 2010/P1/30]

When a charged particle is in a region of magnetic field, which of the following is **not** a necessary condition for the particle to experience a magnetic force?

- A Particle must be charged.
- B Particle must be moving.
- C Particle must be directed at an angle to magnetic flux density.
- D Particle must be accelerating.

3. [Dunman High/H2/Prelims 2010/P1/28]

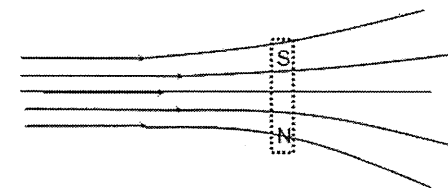
An electron moves in a circular path in a uniform magnetic field of flux density 1.0 mT .

If an α -particle of the same speed is to follow an identical path, what must be the magnitude and direction of the magnetic field? The mass of α -particle is 7200 times the mass of an electron.

	magnitude of magnetic field / mT	direction of magnetic field
A	3600	in opposite direction
B	1800	in opposite direction
C	1800	in same direction
D	3600	in same direction

4. [CJC/H2/Prelims 2010/P1/30]

A bar magnet is to be placed in a non-uniform magnetic field as shown.

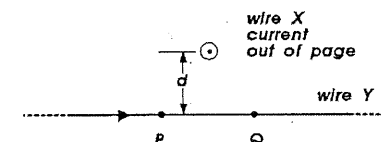


Which line of the table describes the subsequent motion of the magnet?

	rotation	movement
A	anticlockwise	to the left
B	anticlockwise	to the right
C	clockwise	to the left
D	clockwise	to the right

5. [Dunman High/H2/Prelims 2010/P1/29]

Two long straight wires X and Y are placed perpendicular to each other at a distance d apart. A current flows out of the page in wire X. The same current flows from left to right in wire Y.

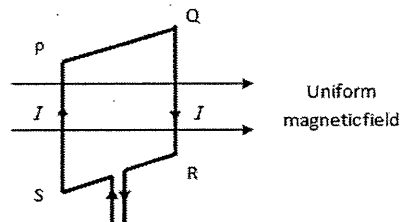


What are the directions of the forces acting on wire Y at points P and Q due to the magnetic field produced by wire X?

	force at P	force at Q
A	out of page	into page
B	into page	out of page
C	towards X	away from X
D	towards X	towards X

6. [HCI/H2/Prelims 2010/P1/28]

A coil PQRS in the shape of a rhombus, has its plane parallel to the magnetic field lines of a uniform magnetic field, as shown.

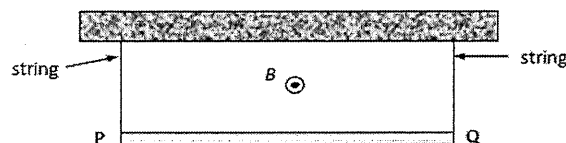


Which of the following statements is true when a current I flows in the coil?

- A There are no forces on sides SP and QR.
- B There are no forces on sides PQ and RS.
- C The direction of force on PQ is acting out of the paper.
- D The magnitude of force on PQ is smaller than the magnitude of the force on QR.

7. [HCI/H2/Prelims 2010/P1/29]

A uniform metallic rod PQ of density ρ , area of cross-section A and length x , is suspended horizontally by two identical strings as shown.



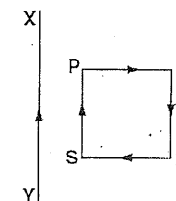
A uniform magnetic field B acts throughout this region in the direction outwards from the plane of this page and perpendicular to the length of the rod. When a certain magnitude of current I is passed along the rod, the tensions of the supporting strings reduce to zero.

What is the magnitude and direction of the current in PQ?

	magnitude	direction
A	$\frac{B}{\rho Ag}$	Q to P
B	$\frac{B}{\rho Ag}$	P to Q
C	$\frac{\rho Ag}{B}$	Q to P
D	$\frac{\rho Ag}{B}$	P to Q

8. [IJC/H2/Prelims 2010/P1/29]

A long straight wire XY lies in the same plane as a square loop of wire PQRS which is free to move. The sides PS and QR are initially parallel to XY. The wire and loop carry steady currents as shown.

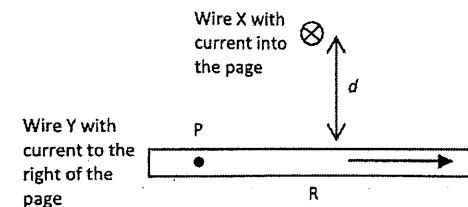


What will be the effect on the loop?

- A It will move towards the long wire.
- B It will move away from the long wire.
- C It will rotate about an axis parallel to XY.
- D It will be unaffected.

9. [MI/H2/Prelims 2010/P1/30]

Two long straight wires, X and Y are placed perpendicular to each other at a small distance, d apart, with wire X vertically above the point R. The current in wire X is into the page and the current in wire Y is to the right as shown.

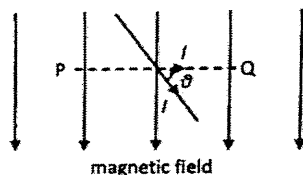


What is the direction of the force acting on wire Y at point P due to the magnetic field produced by wire X?

- A Out of the page
- B Into the page
- C Upwards
- D Downwards

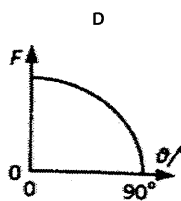
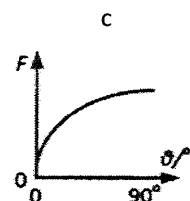
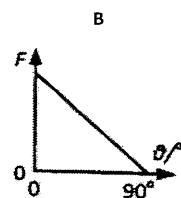
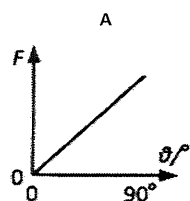
10. [MI/H2/Prelims 2010/P1/29]

A straight wire PQ carrying a constant current I is placed perpendicularly to a uniform magnetic field, as represented by the dotted line in the diagram.



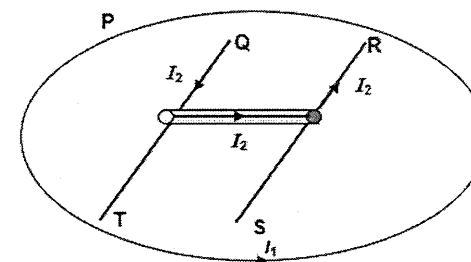
The current-carrying wire is then rotated through an angle θ about an axis perpendicular to the plane of the diagram.

Which of the following graph shows how the magnitude of the magnetic force F on the wire varies with θ in the range 0° to 90° ?



11. [MJC/H2/Prelims 2010/P1/29]

In the diagram below, P is a horizontal circular coil of wire carrying a steady current I_1 . A conducting rod, which is free to move, is supported by 2 fixed horizontal parallel rails TQ and SR which are perpendicular to the length of the conducting rod and carry a constant current I_2 as shown.

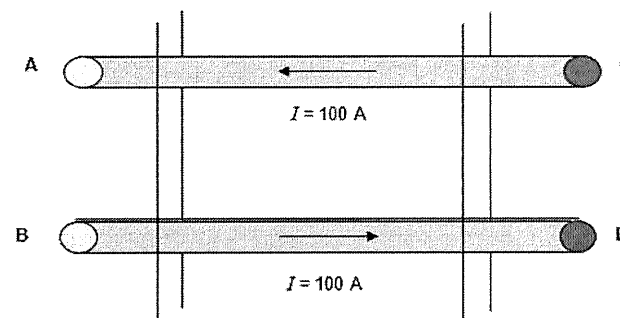


The conducting rod will

- A move towards TS with increasing speed.
- B move towards QR with increasing speed.
- C move towards QR with decreasing speed.
- D be lifted off the horizontal parallel rails momentarily.

12. [MJC/H2/Prelims 2010/P1/30]

The wire AC is free to move vertically while the wire BD is fixed. Both wires carry equal currents of 100 A in opposite directions and that the mass per unit length for each wire is 20.0 g m^{-1} .

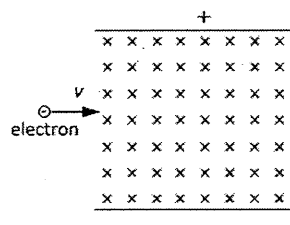


What is the height of wire AC above wire BD ? (You may assume that the magnetic field, B at a distance r from a wire carrying current I to be $B = \frac{\mu_0 I}{2\pi r}$.)

- A zero, wires AC and BD are attracted to each other.
- B 0.102 cm
- C 1.02 cm
- D 10.2 cm

13. [NJC/H2/Prelims 2010/P1/30]

An electron enters a region where there are uniform electric field and magnetic field of 0.50 T acting at right angles to its horizontal velocity v , as shown in the figure below. The electron remained undeflected when it enters the region. When the magnetic field is switched off, the electron immediately moved a vertical displacement of 0.15 mm in 1.0 ns.

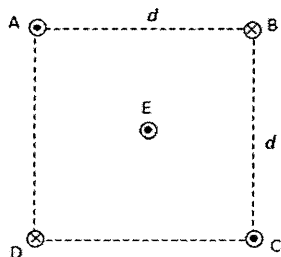


What is the speed v of the electron?

- A $0.85 \times 10^3 \text{ m s}^{-1}$ B $3.4 \times 10^3 \text{ m s}^{-1}$ C $150 \times 10^3 \text{ m s}^{-1}$ D None of the above

14. [NJC/H2/Prelims 2010/P1/31]

Five straight and parallel wires are arranged as shown in the diagram below. Wire A, C and D carry a steady current I . Wire B and E (at centre of wire) carry a steady current $2I$.



Which of the following statements about the wire at E is correct?

- A It does not experience any force.
B It experiences a force in the direction of A
C It experiences a force in the direction of B.
D It experiences a force in the direction of D.

15. [NYJC/H2/Prelims 2010/P1/28]

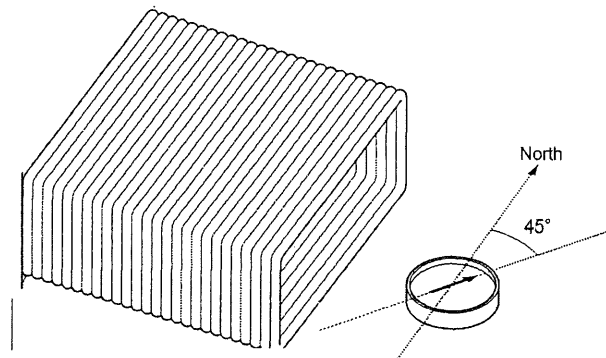
A bubble chamber photograph shows a proton moved in a circular arc with radius 10 cm and perpendicular to a magnetic field of flux density 0.30 T.

What is the kinetic energy of the proton?

- A 4.0 keV B 40 keV C 400 keV D 4.0 MeV

16. [NYJC/H2/Prelims 2010/P1/29]

The diagram shows a compass near a rectangular coil of wire.



A direct current flows in the coil creating a magnetic field. The field has caused the compass to be deflected away from the north and towards the east. The angle of deflection is about 45° .

The current in the coil is doubled. Which one is the approximate value of the new angle of deflection?

- A 20° B 60° C 90° D 135°

17. [PJC/H2/Prelims 2010/P1/30]

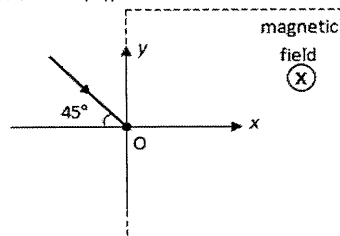
An electron is moving along the axis of a solenoid carrying a current.

Which statement about the electromagnetic force acting on the electron is correct?

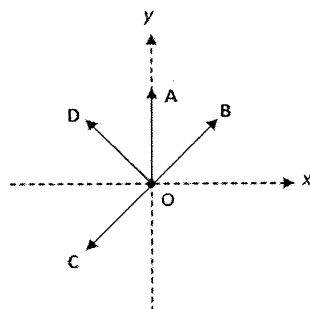
- A The force acts radially inwards.
B The force acts radially outwards.
C The force acts in the direction of motion.
D No force acts.

18. [PJC/H2/Prelims 2010/P1/29]

An electron enters a region at point O at an angle of 45° to the x-axis as shown. The region contains a magnetic field which is directed into the plane of the page.

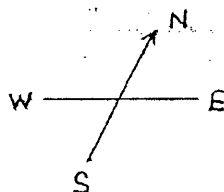
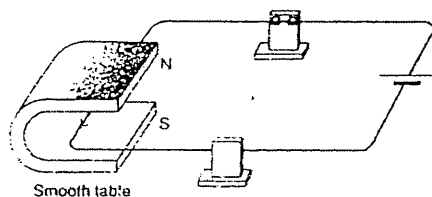


An electric field is applied such that the electron passes through the region undeflected. Which of the following shows the direction of the electric field from O?



19. [VJC/H2/Prelims 2010/P1/30]

A small horse shoe magnet is placed on a smooth horizontal table. A wire carrying a current is inserted into the space between the poles of a magnet as shown in the figure. The wire is held fixed by two insulating stands on the ground.

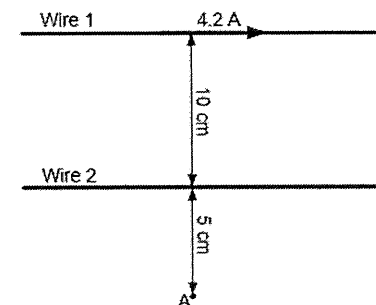


What happens to the magnet?

- A It remains stationary.
- B It accelerates towards the North.
- C It accelerates towards the East.
- D It accelerates towards the West.

20. [RI/H2/Prelims 2010/P1/29]

Wire 1 carries a current of 4.2 A to the right as shown.

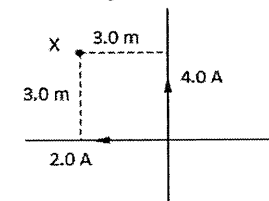


What is the magnitude and direction of the current that is carried in Wire 2 so that the net magnetic flux density at point A is $4.0 \times 10^{-6} \text{ T}$?

- A 0.40 A to the left
- B 0.40 A to the right
- C 1.4 A to the left
- D 1.4 A to the right

21. [RVHS/H2/Prelims 2010/P1/28]

Two long current carrying conductors are placed perpendicular to each other. The current flowing through one of the wires is 4.0 A upwards, while the current through the other wire is 2.0 A towards the left.



What is the magnitude and direction of the resultant magnetic field at a point X, which is 3.0 m perpendicularly away from both wires? Ignore the Earth's magnetic field.

- A $1.33 \times 10^{-7} \text{ T}$ out of the plane of the page
- B $4.00 \times 10^{-7} \text{ T}$ into the plane of the page
- C $4.00 \times 10^{-7} \text{ T}$ out of the plane of the page
- D $2.67 \times 10^{-7} \text{ T}$ into the plane of the page

22. [RI/H2/Prelims 2010/P1/30]

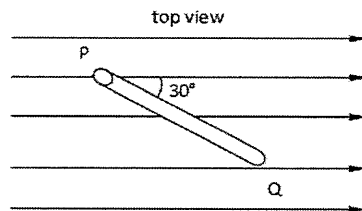
One end of a flat rectangular coil of negligible mass is placed at the centre of a 1000-turn circular coil of diameter 25 cm as shown in the figure below. A current of 5.0 A is passed through the rectangular coil and when a 5.0 g paper rider is placed at 2.0 cm to the right of the pivot, the rectangular coil is balanced horizontally.

What is the magnitude of the current that the 1000-turn circular coil must carry in order for the rectangular coil to remain horizontal?

- A 3.3 A B 5.0 A C 6.5 A D 9.0 A

23. [RVHS/H2/Prelims 2010/P1/29]

A horizontal wire PQ of length 0.50 m and weight 0.50 N is placed at an angle 30° to the magnetic field as shown below. The wire is balanced by the magnetic force of the magnetic field of magnetic flux density of 1.0 T.

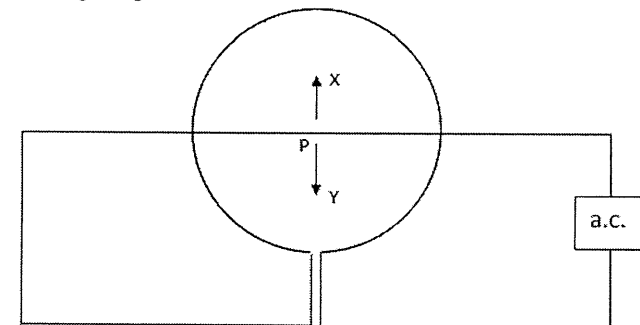


What is the magnitude and direction of the current in the wire?

	magnitude	direction
A	1.2 A	P to Q
B	2.0 A	P to Q
C	1.2 A	Q to P
D	2.0 A	Q to P

24. [SAJC/H2/Prelims 2010/P1/23]

An insulated wire is bent into a circular coil and placed above a straight portion as shown. The terminals of the wire are connected to an alternating voltage.

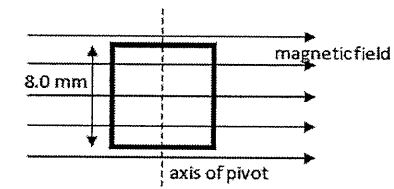


What is the direction of the force acting at point P, the centre of the circle, on the wire?

- A It is oscillating in the X and Y direction.
 B It is pointing in the X-direction.
 C It is pointing in the Y-direction.
 D It is oscillating into and out of the page.

25. [TJC/H2/Prelims 2010/P1/29]

A 20-turn square coil of side 8.0 mm is pivoted at the centre and placed in a magnetic field of flux density 0.010 T such that two sides of the coil are parallel to the field and two sides are perpendicular to the field, as shown in the figure. A current of 5.0 mA is passed through the coil.

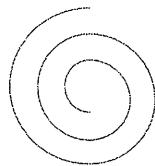


What is the torque created on the coil?

- A 1.6×10^{-9} N m B 3.2×10^{-8} N m C 6.4×10^{-8} N m D 3.2×10^{-5} N m

26. [TJC/H2/Prelims 2010/P1/30]

An electron is moving in air at right angles to a uniform magnetic field. The diagram shows the path of the electron. The electron is slowing down.

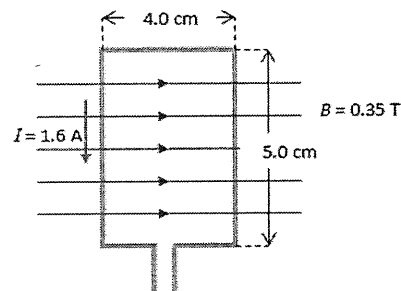


What is the direction of motion of the electron and the direction of the magnetic field?

	direction of motion	direction of magnetic field
A	clockwise	into plane of paper
B	clockwise	out of plane of paper
C	anti-clockwise	into plane of paper
D	anti-clockwise	out of plane of paper

27. [YJC/H2/Prelims 2010/P1/29]

A rectangular coil with 200 turns has dimensions 5.0 cm by 4.0 cm. The coil is placed in a magnetic field B of 0.35 T and the current I in the coil is 1.6 A as shown.



What is the value and direction of the maximum torque?

- A 0.22 N m with the left edge moving into plane of paper
 B 5.6 N m with right edge moving into plane of paper
 C 0.22 N m with right edge moving into plane of paper
 D 11.2 N m with left edge moving into plane of paper

II. Structured Questions

1. [CJC/H2/Prelims 2010/P3/1]

- a. State what is meant by a *field of force*. [1]
- b. Atoms of Neon-20 are ionised by the removal of one electron from each atom. These ions are accelerated through a potential difference of 1400 V. They are then injected into a region of space where there are uniform electric and magnetic fields acting in right angles to the original direction of motion of the ions. The region of magnetic and electric fields and polarities of the electric plates are shown in Fig. 1.1. (Hint: The mass of 1 atom of Neon-20 is $20u$, where u is the unified atomic mass constant and has a value of 1.66×10^{-27} kg.)

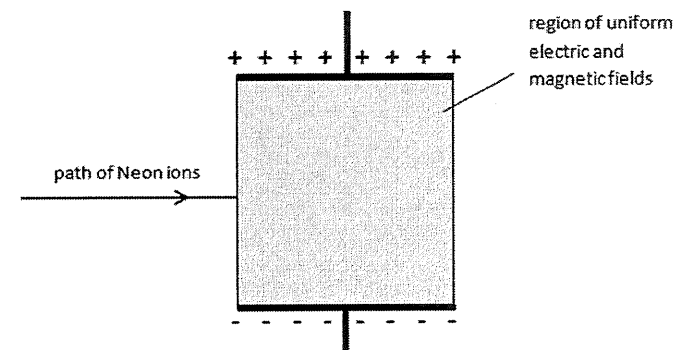


Fig. 1.1

The electric field has field strength E and the flux density of the magnetic field is B .

- i. State the charge of a Neon ion. [1]
- ii. Calculate the speed of the Neon ions before they enter into the region of the electric and magnetic fields. [3]
- iii. 1. State the direction of the electric force acting on a Neon ion when it enters into the shaded region. [1]
2. Draw the direction of the magnetic field onto the shaded region in Fig. 1.1 such that the Neon ions pass through the shaded region undeflected [1]
- iv. The electric field strength E is 6.2×10^3 V m⁻¹. Calculate the magnitude of the magnetic flux density so that the ions are not deflected in the region of the fields. [2]
- c. The mechanism by which the Neon atoms in b. are ionised is changed so that each atom loses two electrons instead of just one. State the changes that occur in
- i. the speed of the ions entering the region of the electric and magnetic fields in b. [1]
- ii. the path of the ions in the two fields. [1]

[Total: 11]

2. [CJC/H2/Prelims 2010/P3/6 (part)]

- a. Define *magnetic flux density* and *magnetic flux linkage*. [3]
- b. A wire with resistance $8.66 \, \Omega$ per metre and diameter $0.050 \, \text{mm}$ is used to construct an electromagnet in the form of a hollow solenoid by winding one layer of close-packed turns onto a plastic tube of length $200 \, \text{mm}$ and diameter $30 \, \text{mm}$. The solenoid is connected in series with a switch and a battery of e.m.f. $6.0 \, \text{V}$ and negligible resistance, as shown in Fig. 2.1.

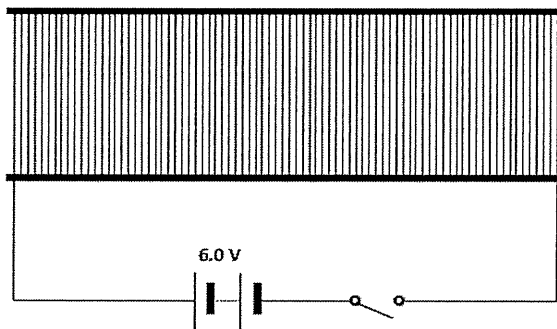


Fig. 2.1

When the switch is turned on, the magnetic flux density B produced at the centre of the solenoid is given by

$$B = \mu_0 n I,$$

where n = number of turns per unit length.

- i. Calculate the number of turns of the solenoid. [1]
 - ii. Calculate the resistance of the wire of the solenoid. [2]
 - iii. Calculate the maximum magnetic flux density produced by the electromagnet. [3]
- c. State and explain the effects on the magnetic flux density of each of the following changes.
- i. The plastic tube is filled with iron filling. [3]
 - ii. Twice the length of similar wire is used so that the coil consists of two close-packed layers. [3]

[Total: 15]

3. [Dunman High/H2/Prelims 2010/P2/4]

In Fig. 3.1, a uniform magnetic field B pointing into the paper and a uniform electric field E pointing upwards are applied on the left-hand side and right-hand side of the y -axis respectively.

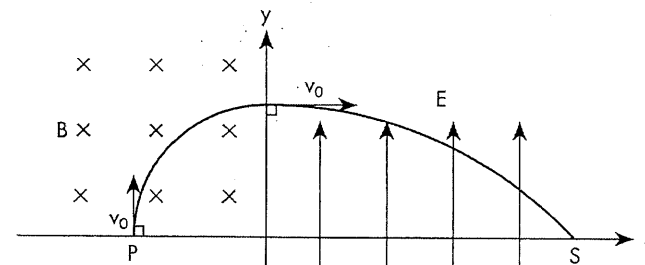


Fig. 3.1

A particle of mass m and charge q enters the magnetic field with an initial velocity v_0 at point P. After leaving the magnetic field, it moves into the electric field and leaves at point S. Neglect gravitational effects.

- a. i. State the sign of the charge carried by the particle. [1]
- ii. Describe the motion of the particle in the two different fields. [2]
- b. Deduce an expression, in terms of m , q , B , E or v_0 , where appropriate, for the time taken for the particle to travel through
 - i. the magnetic field [3]
 - ii. the electric field. [3]
- c. If the velocity of the particle at point S is v , deduce an expression, in terms of m , v_0 and v , for the work done on the particle by the field as the particle travel through
 - i. the magnetic field [1]
 - ii. the electric field. [1]

[Total: 11]

4. [HCl/H2/Prelims 2010/P3/6 (modified)]

a. Define *magnetic flux density*.

[2]

b. i. The magnetic flux density B a distance d from a long, straight wire carrying a current I is given by

$$B = \frac{\mu_0 I}{2\pi d}$$

Using this relationship with a clearly labeled diagram, show that the force per unit length between two long, straight, parallel wires carrying currents I_1 and I_2 respectively in the same direction, separated by a distance d is given by

$$\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d} \quad [2]$$

ii. A light metal spring with its axis vertical is clamped at the top and is hanging freely unextended. The turns of the spring have a radius r and the longitudinal distance between adjacent turns is d when the spring is unstretched, as shown in Fig. 4.1. The spring obeys Hooke's Law and has a spring constant k .

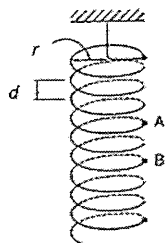


Fig. 4.1

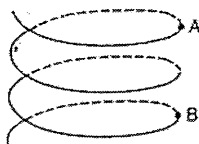


Fig. 4.2

An e.m.f. source is then connected across points A and B, and a current I passed through exactly two turns of the spring.

1. On Fig. 4.2, complete the circuit connections made at A and B so that the current passing through the spring can be measured. [2]

2. A student noted that the current causes a change in the length of the spring. State and explain whether the change in the length of the spring is an extension or compression.

(Assume the connections themselves are light, the equilibrium of the spring is not disturbed, and the force between any two circular turns of the spring is the same as that between two straight wires.) [3]

3. The connections made across A and B are now reversed.

State whether the change in the length of the spring is an extension or a compression. [1]

c. (*Note: concepts from **Electromagnetic Induction** are required to solve this part of the question.)

A physicist designs an electromagnetic braking system for a truck in his model railway. The top view of the setup is shown in Fig. 4.3.

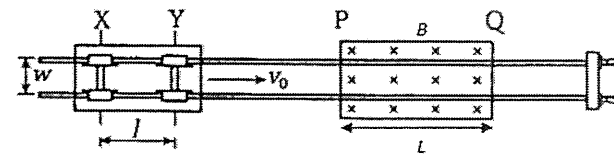


Fig. 4.3 (top view)

The truck has mass $M = 0.15$ kg. X and Y are wheel axles with an axle spacing of $l = 0.10$ m. The truck is projected at velocity v to the right, along a level metal track with rail spacing $w = 0.050$ m. Mechanical friction is negligible. The truck runs into a limited region of vertical magnetic field $B = 0.70$ T over the length $L = 0.15$ m between points P and Q. The field is negligible outside this region. The truck has metal wheels and axles. Axles X and Y, together with the section of rail length l between the axles, form a closed loop. The closed loop has a circuit resistance R of 0.020Ω .

When the initial velocity $v = 30 \text{ mm s}^{-1}$, the braking system works well with the truck coming to rest before axle X enters the magnetic field at point P.

i. Explain how the braking occurs. [3]

ii. Hence, by considering the net horizontal force, show that the acceleration, a , experienced by the truck when it enters the magnetic field is proportional to its velocity v . [3]

iii. Suggest and explain briefly one improvement to this braking system to effectively stop trucks projected at high velocities. [2]

[Total: 18]

5. [HCI/H2/Prelims 2010/P3/7]

- a. List two similarities and two differences between electric field and magnetic field. [2]
- b. Fig. 5.1 shows a cyclotron, which is used for accelerating charged particles to very high kinetic energies. It consists of two hollow semi-circular metal chambers, called 'dees' of radius 1.30 m, within which there is a uniform magnetic field, B . A high frequency square wave alternating voltage V and frequency 3.05 MHz is connected across the narrow gap between the chambers.

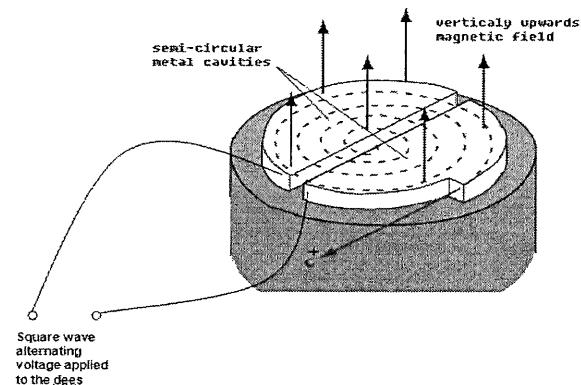


Fig. 5.1

Fig. 5.2 shows an enlarged view of the narrow gap at the centre of the cyclotron. Positive charges called deuterons enter the narrow gap near dee Y with negligible kinetic energy (point 1). They accelerate towards dee X, which has lower potential (point 2).

There is no electric field within the dees and the deuterons move in semi-circular path inside the dee X. As soon as they complete a semi-circular path (point 3), dee X reverses its polarity such that dee Y now has a lower potential. The deuterons then accelerate towards dee Y (point 4).

This process is repeated until the deuterons have gained sufficient energy, eventually emerging at a very high speed.

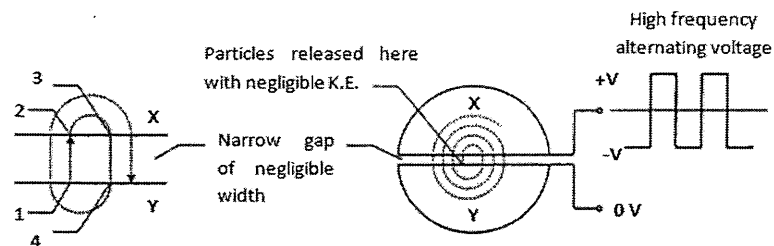


Fig. 5.2

- i. Explain why the deuterons move in a semicircular path when they are inside the dees. [2]

- ii. Show that the radius of the circular path of the deuteron motion inside a dee is given by

$$r = \frac{mv}{Bq},$$

where m is the mass of the deuteron, v is its speed and q is its charge and B is the magnetic flux density inside the chambers. [2]

- iii. Show that the period of the motion is independent of v and r . [2]
- iv. Sketch on the axes in Fig. 5.3 to show the variation of the kinetic energy K.E. of the deuteron with distance travelled as it moves from Point 2 to Point 4. [2]

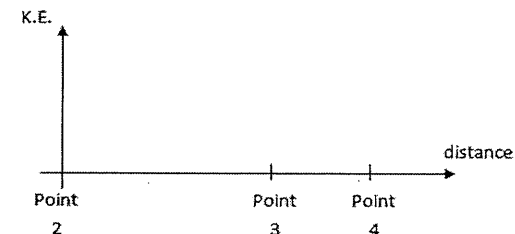


Fig. 5.3

- v. There is a uniform magnetic field of 0.400 T directed normally out of the plane of the dees and the high frequency alternating voltage V of amplitude 100 V and frequency 3.05 MHz is connected between the dees.

The mass of deuteron is 3.34×10^{-27} kg and its charge is $+1.60 \times 10^{-19}$ C.

- Calculate the gain in kinetic energy of the deuterons in each complete revolution. [2]
- Show why a frequency of 3.05 MHz for the square wave alternating voltage (in Fig. 5.1) is appropriate. [3]
- Hence, calculate the time taken for a deuteron to attain a kinetic energy of 1.00 MeV. [3]

[Total: 20]

6. [JC/H2/Prelims 2010/P2/4]

a. Define

i. *magnetic flux density* [1]ii. the *tesla*. (not in syllabus) [1]

- b. Fig. 6.1 shows an electric motor which is made up of a rectangular coil of wire of 150 turns. The coil is 0.20 m long and 0.12 m wide. The coil has a current of 0.32 A flowing through it and its plane is parallel to a field of magnetic flux density 0.36 T.

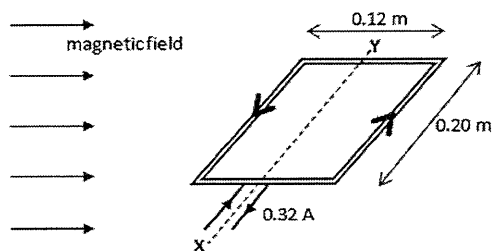


Fig. 6.1

- i. Draw arrows on Fig. 6.1 to represent the directions of the magnetic forces acting on the coil. Label them F . [1]
- ii. Calculate the magnitude of the magnetic force acting on one side of the coil. [2]
- iii. Calculate the torque which is exerted on the coil. [1]

[Total: 6]

7. [JC/H2/Prelims 2010/P3/3]

- a. A charged particle may experience a force in an electric field and in a magnetic field.

State two differences between the forces experienced in the two types of field. [2]

- b. A proton, travelling in a vacuum at a speed of $4.5 \times 10^6 \text{ m s}^{-1}$, enters a region of uniform magnetic field of flux density 0.12 T. The path of the proton in the field is a circular arc, as illustrated in Fig. 7.1.

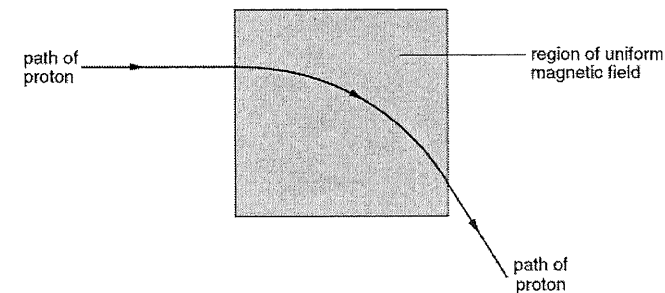


Fig. 7.1

- i. State the direction of the magnetic field. [1]
- ii. Calculate the radius of the path of the proton, in terms of cm, in the magnetic field. [3]
- c. A uniform electric field is now created in the same region as the magnetic field in Fig. 7.1, so that the proton passes undeflected through the region of the two fields.
- i. On Fig. 7.1, draw an arrow to show the direction of the electric field and label the arrow E . [1]
- ii. Calculate the magnitude of the electric field strength. [2]
- d. Suggest why gravitational force on the proton has not been considered in the calculations in b. and c.. [1]

[Total: 10]

8. [MI/H2/Prelims 2010/P2/5]

- a. Define *magnetic flux density*. [2]
- b. Fig. 8.1 shows a loosely-coiled metal spring hanging from a fixed point.

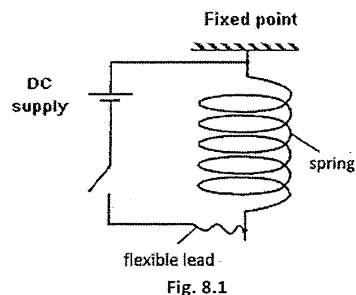


Fig. 8.1

The ends of the spring are connected to a circuit as shown in Fig. 8.1. When a current is switched on, the spring becomes compressed (i.e. the coils move closer to one another and the vertical length of the spring becomes slightly shorter than before).

- i. Explain why the spring shortens in length when the current is switched on. [2]
- ii. State and explain whether the spring is lengthened or compressed when the direction of current is reversed. [1]
- c. Fig. 8.2 shows a beam of α -particles (which is positively charged) of speed v entering a region in space where there is a uniform magnetic field of flux density B .

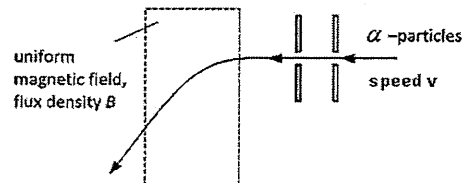


Fig. 8.2

In the magnetic field, they are deflected from their original path and move along the arc of a circle of radius r .

- i. On Fig. 8.2, indicate the direction of the magnetic field. [1]
- ii. Sketch the path of α -particles that has a speed greater than v . [1]
- d. A beam of electrons that is travelling with velocity $3.4 \times 10^7 \text{ m s}^{-1}$ enters perpendicularly into a region with magnetic flux density of 8.4 mT .
- i. Calculate the magnitude of the magnetic force on an electron. [2]
- ii. State and explain the changes (if any) in the kinetic energy of electron due to the magnetic force acting on it. [1]

[Total: 10]

9. [MJC/H2/Prelims 2010/P3/7(part)]

- a. Fig. 9.1 shows an arrangement used to accelerate an initially stationary alpha particle (which is positively charged) and make it travel in a uniform magnetic field.

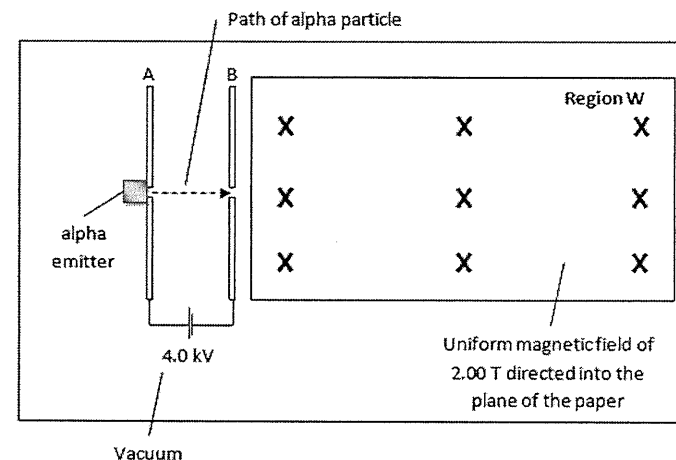


Fig. 9.1

- i. On Fig. 9.1, draw a possible trajectory of the alpha particle in the uniform magnetic field. [1]
- ii. Explain whether the force experienced by the alpha particle due to the magnetic field changes its kinetic energy. [3]
- iii. Show that the alpha particle will attain a speed of $6.21 \times 10^5 \text{ m s}^{-1}$ when it reaches the slit opening of plate B. (mass of alpha particle = $6.644 \times 10^{-27} \text{ kg}$) [1]
- iv. Calculate the electric field that needs to be applied in Region W for the alpha particle to pass through the uniform magnetic field undeflected. State its direction. [3]
- b. The arrangement in Fig. 9.1 now modified so that the alpha particle enters the uniform magnetic field at an angle of 30° to the horizontal as shown in Fig. 9.2.

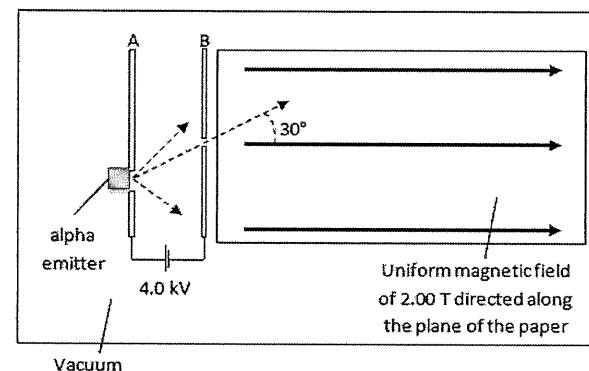


Fig. 9.2

The motion of the alpha particle can be described as a helix as shown in Fig. 9.3.

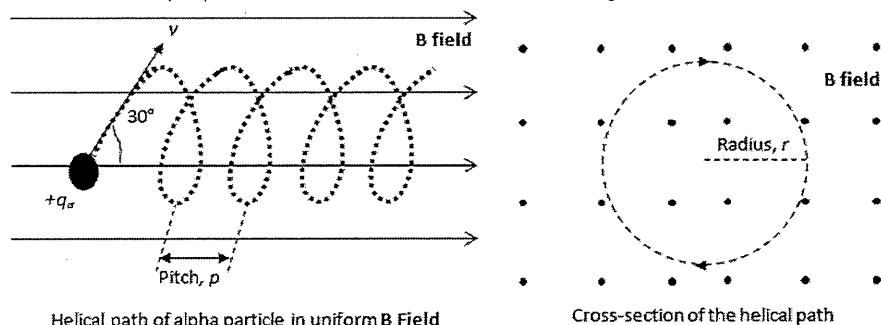


Fig. 9.3

- Calculate the radius r of the helical path.
- Show that the period T of the helical path can be expressed as

$$T = \frac{2\pi m_\alpha}{q_\alpha B}$$

where m_α and q_α is the mass and charge of the alpha particle respectively.

- Calculate the pitch, p .
- Describe and explain how the helical path will change if a positron, ${}_{+1}^0e^+$ (i.e. a particle with the mass of an electron and charge $+1.6 \times 10^{-19}$ C) with the same initial velocity was to be used in the experiment instead. You may quote relevant equations to substantiate your answer.

[Total: 18]

10. [PJC/H2/Prelims 2010/P2/4]

- Explain what is meant by a *field of force*.
- Describe, by means of a well-labelled diagram, the motion of electrons initially moving at right angles in a uniform
 - electric field.
 - magnetic field.
- A rectangular strip of copper of dimensions $10 \text{ mm} \times 5 \text{ mm} \times 1 \text{ mm}$ carries a conventional current I . A magnetic field B of flux density 1.0 T is applied in a direction perpendicular to the strip as shown in Fig. 10.1.

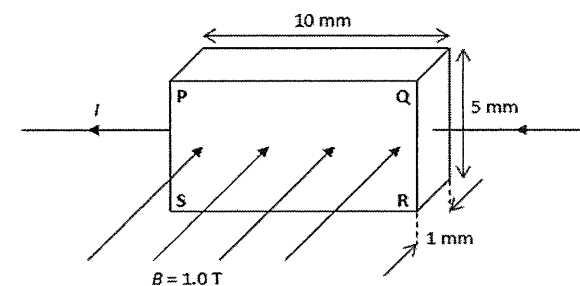


Fig. 10.1

- Explain how a voltage is set up across side PQ with respect to SR.
- Calculate the voltage, given that the velocity of the electrons is $2.52 \times 10^{-5} \text{ m s}^{-1}$.

[Total: 10]

11. [RI/H2/Prelims 2010/P3/4]

A cyclotron is a device used to accelerate ions to very high speeds. Fig. 11.1 shows a diagram of a cyclotron viewed from above. It is composed of two hollow, semi-circular electrodes called “Dees”. The “Dees” are encased inside a vacuum chamber and exposed to a perpendicular uniform magnetic field. An ion source lies in between the “Dees” at point A. An alternating voltage supply is connected across the “Dees”.

During operation, the voltage supply produces an alternating electric field in the small gap between the “Dees”. This is to ensure that the ions are accelerated each time they cross the gap. On entering the “Dees”, the uniform magnetic field causes the ions to move in a circular path. As the ions speed up, they travel in ever larger circles within the “Dees”. Once the ions reach a sufficiently large speed, they exit through an outlet in one of the “Dees” which is aimed at a target.

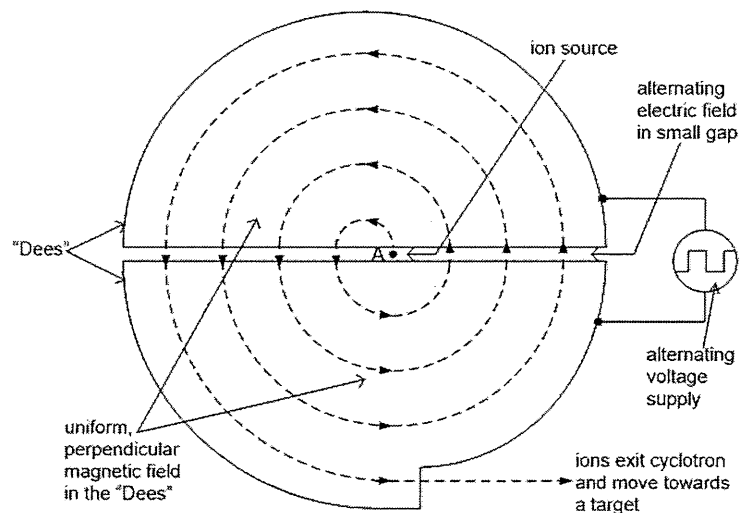


Fig. 11.1

At any time when an ion of mass m and charge q accelerates across the small gap, the potential difference between the “Dees” is V . The ion then travels in a circular path in the “Dees” where a uniform magnetic field of flux density B is applied perpendicularly.

- Show that the time T for the ion to complete one revolution is $\frac{2\pi m}{Bq}$. Ignore relativistic effects. [2]
- A helium nucleus of mass 6.68×10^{-27} kg and charge $2e$ is accelerated in the cyclotron by applying an alternating potential difference of 450 V across the “Dees”. The magnetic flux density through the “Dees” is 0.850 T.
 - Calculate the time T to complete one revolution for the helium nucleus. [2]
 - Determine the frequency f of the alternating voltage supply so that the helium nucleus is accelerated every time it crosses the gap between the “Dees”. [2]
 - State an expression for the gain in kinetic energy of the helium nucleus after one revolution in terms of e and V . [1]
 - Hence, determine the speed v of the helium nucleus after five revolutions. [3]

[Total: 10]

12. [YJC/H2/Prelims 2010/P2/3]

A narrow beam of electrons at a speed of 3.2×10^7 m s⁻¹ travels along a circular path in a uniform magnetic field of flux density, B , as shown in Fig. 12.1.

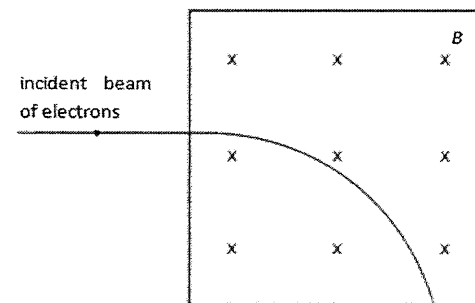


Fig. 12.1

- Explain why the electrons undergo uniform circular motion. [3]
- Show that the speed v of the electrons in the field is given by
$$v = \frac{Ber}{m}$$
 where r is the radius of the circular path of the beam in the field. [2]
- The radius of the circular path of the beam in the field was found to be 25 mm. Determine the magnetic flux density of the field. [2]

[Total: 7]

13. [SAJC/H2/Prelims 2010/P3/7]

- a. Electrons from a filament source enter a region between the parallel plates after being accelerated by an electric field. Fig. 13.1 shows the electrons travelling horizontally at a speed of $2.50 \times 10^7 \text{ m s}^{-1}$ entering the pair of parallel plates.

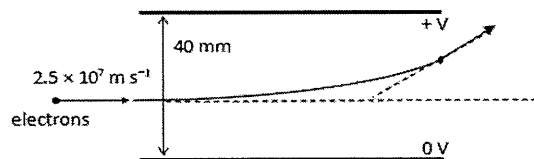


Fig. 13.1

- The electrons deviate by 30° on leaving the parallel plate of 80.0 mm long. The separation between the plates is 40.0 mm.
Calculate the time taken for the electrons to traveling through the plates. [1]
 - Calculate the vertical component of the velocity when the electrons exit the parallel plates. [2]
 - Hence, calculate the acceleration of the electrons. [1]
 - Calculate the p.d. V between the two plates. [2]
- b. A magnetic field is applied to the region of the electric field in Fig. 13.1 so that the electrons pass straight through undeflected.
- Indicate in words or sketch the direction of the magnetic flux density in Fig. 13.1. [1]
 - Calculate the magnetic flux density required to produce this effect. [2]
- c. The parallel plates in Fig. 13.1 are removed but the magnetic field is kept at the same value and direction.
- Draw the path taken by the electrons in the magnetic field. [1]
 - Calculate the radius of curvature of the path taken by these electrons. [2]
 - Determine the speed of the electrons upon leaving the field. Explain your reasoning. [2]
- d. (*Note: concepts from Quantum Physics are required to solve this part of the question)
- The magnetic field in part c. above is removed. The speed of the electron is reduced and the rate of electrons reduced to one per second. The electrons are allowed to hit a microgrid with a vacuum-deposited gold film. It was found that high quality interference patterns are observed on photographic films placed beyond the microgrid.
- Explain how the electrons can produce the interference pattern even when there is only one electron per second approaching the grid. [3]
 - State how the interference pattern can be made wider by changing the properties of the electrons only. [1]

[Total: 20]

14. [TJC/H2/Prelims 2010/P2/3]

- a. Define magnetic flux density. [1]
- b. A beam of electrons is generated inside an evacuated flask which is placed in a magnetic field of flux density $6.5 \times 10^{-4} \text{ T}$. The electrons are travelling at right angles to the field, which is directed into the paper, as shown in Fig. 14.1. The arrow indicates the initial direction of the electron beam.

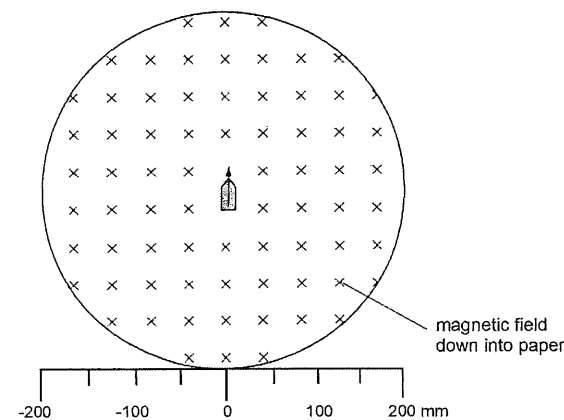


Fig. 14.1

- The electrons are observed to travel in a circle of radius 83 mm.
Calculate the speed of the electrons. [3]
- On Fig. 14.1, draw a possible path of the electrons. [1]
- The initial direction of the electron beam is then changed so that it makes an angle of 80° to the magnetic field. The electrons then travel in a helical path, as illustrated in Fig. 14.2.

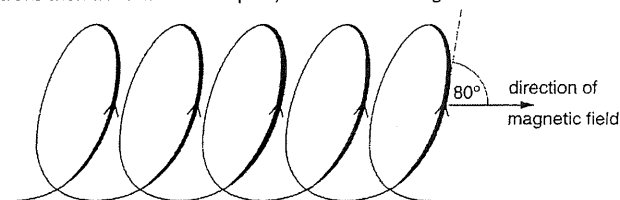


Fig. 14.2

The helical path appears to be circular when viewed along the direction of the magnetic field but the electrons have a constant velocity along the field.

Explain why the electrons follow a helical path.

[2]
[Total: 7]

15. [VJC/H2/Prelims 2010/P3/5]

- a. i. Define *magnetic flux density* and state its SI unit. [2]
 ii. Express the unit of magnetic flux density in terms of its base units. [4]
- b. Fig. 15.1 shows the arrangement of a mass spectrometer, which is an instrument to measure the masses of ions. An ion of mass m and charge $+q$ is produced at rest in an ion source S . The ion is accelerated through a pair of parallel plates P_1 and P_2 of potential difference V . The ion is then allowed to enter a region of uniform magnetic field of flux density B , which is directed perpendicularly to the plane of the paper. In the field it moves in a semicircle, striking a photographic plate at a distance x from the entry slit.

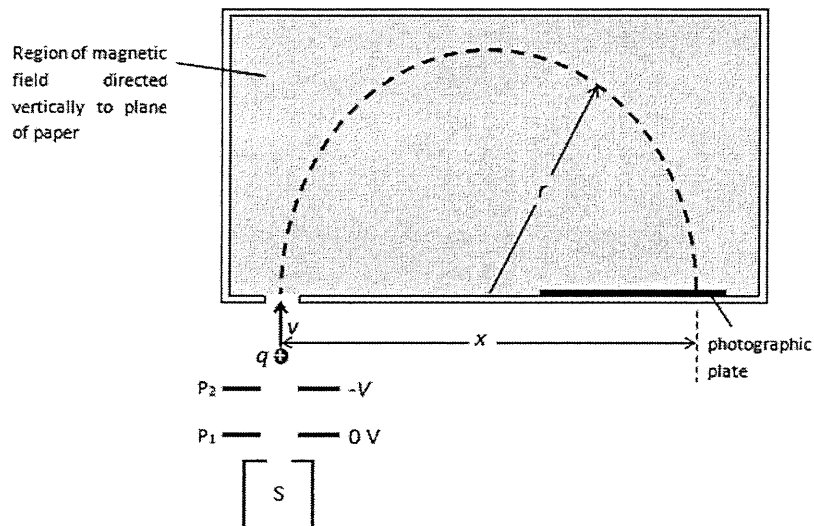


Fig. 15.1

- i. Show that the ions exit from plate P_2 and enter the magnetic field with a speed $v = \sqrt{\frac{2qV}{m}}$. [1]
- ii. Describe and explain the effects of the magnetic field on the velocity of the ion upon its entry into the magnetic field. [2]
- iii. If the ion is positively-charged, state the direction of the magnetic field. [1]
- iv. For an ion that enters the magnetic field with velocity v , show that the radius r of its semi-circular path is $r = \frac{mv}{Bq}$. [2]
- v. Hence find an expression for x in terms of m , B , q and the potential difference V of the parallel plates. [2]
- c. i. ^{12}C and ^{14}C are isotopes of carbon. Ions of these two isotopes were introduced into the mass spectrometer described in b. and two spots were observed on the photographic plate. Explain why there would be two spots. [2]

- ii. The singly and positively charged ions from the two isotopes were accelerated through a potential difference V of 4.0 kV and enter the magnetic field of flux density B of 0.50 T. If the masses of the ions are 12 u and 14 u respectively, calculate the distance Δx between the spots on the photographic plate. [4]
- d. If an electron were to be introduced into the mass spectrometer, describe and explain what would be observed if the accelerating potential and the magnetic field remained unchanged. (Assume that the polarity of the plates P_1 and P_2 are reversed.) [3]

[Total: 20]

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
D	D	A	A	B	D	C	A	A	D	A	C	B	D	B

16	17	18	19	20	21	22	23	24	25	26	27
B	D	C	D	A	A	C	B	C	C	D	C

- D** Straight wire produces magnetic field into paper at point P (using Right-hand grip rule for straight wire). Then using Fleming's Left hand rule at point P, First finger (B-field) into paper, Middle finger (current) tangential in the direction of arrow A, then the thumb (motion or force) will naturally point in the direction of arrow D.
- D** Charged particle experiences magnetic force as long as it is in motion, whether it is moving at constant speed or accelerating.
- A** Circular motion : $Bqv = mv^2/r$, hence Bq is proportional to m (since v and r is constant in this question)
 Charge of α -particle is $+2e$, charge of electron is $-e$
 mass of α -particle = 7200 mass of electron,
 $B_2q_2 / m_2 = B_1q_1 / m_1$
 $B_2(2) / 7200 = (1)(1) / (1)$
 $B_2 = 3600 \text{ mT}$
 Since α -particle charge is of opposite sign as electron, the B-field has to be in opposite direction in order for the α -particle to move in same direction as the electron.
- A** The direction of the external B-field is left to right, hence can image the B-field is created by a N-pole on the left, S-pole on the right. Since the field lines are denser on the left, it suggests the N-pole on the left has a stronger magnetic flux density. Hence the magnet will move to the left due to the stronger attraction to the left.
 As it moves, it will also rotate anticlockwise as the magnet's S-pole is attracted to the external magnetic N-pole (and vice-versa).

- B** Using right-hand grip rule for the wire X, the B-field due to wire X is anti-clockwise on the plane of paper cutting through point P and Q, giving a vertical component downwards at point P, and vertical component pointing upwards at point Q.
 Using Fleming's Left Hand rule at point P and Q, the direction of magnetic force can be obtained at points P and Q.
- D** Option A and B: Since none of the sides are parallel to the B-field, all sides will experience a magnetic force.
 Option C: Direction of force is into paper.
 Option D: force on QR is BIL while force on PQ is $BIL \sin \theta$, where θ is the angle between B-field normal to the side PQ.
- C** Weight of rod balances the magnetic force acting upwards. Direction of current has to be from Q to P such that magnetic force is upwards (using FLHR).

$$mg = BIL \Rightarrow \rho A x g = BIL \Rightarrow I = \frac{\rho Ag}{B}$$
- A** Like currents attract. PS attracted to XY. Although QR is repelled by XY, it is further away and hence the repulsive force is weaker.
- A** Using Right-hand grip rule, the B-field due to Wire X is at an angle at point P, which can be resolved into an upward and leftward components. The upward component will cause a magnetic force at point P. Using FLHR, the force is out of page.
- D** Magnetic force = $BIL \cos \vartheta$. When $\vartheta = 0$, force is maximum as conductor is perpendicular to the B-field. Note that when the expression $F = BIL \sin \theta$ is used, the angle θ is the angle the conductor makes with the B-field. Hence in this case, the angle ϑ is actually = $90^\circ - \theta$, hence cosine is used instead.
- A** P produces a B-field upwards. Using FLHR, the rod will experience a magnetic force towards TS.
- C** Consider wire AC : $mg = BIL$

$$\left(\frac{m}{L}\right)g = BI$$

$$(0.020)(9.81) = \frac{(4\pi \times 10^{-7})(100)}{2\pi r} (100)$$

$$r = 0.0102 \text{ m}$$
- B** Initially, $Bqv = qE \Rightarrow v = E/B$ ———(1)
 Without B-field, E-force on electron = $qE = ma \Rightarrow a = qE/m$
 Using $s = ut + \frac{1}{2}at^2$,
 $0.00015 = \frac{1}{2}(qE/m)(1 \times 10^{-9})^2$
 $E = 1708 \text{ N m}^{-1}$ (where mass of electron : $9.11 \times 10^{-31} \text{ kg}$; charge of electron : $1.60 \times 10^{-19} \text{ C}$)
 Substitute into (1), $v = 3.4 \times 10^3 \text{ m s}^{-1}$

- 14 D Like currents attract; unlike currents repel. Force due to A and C acting on E will cancel each other. Force due to B is stronger (than that from D), hence will repel E towards D.

15 B $Bqv = mv^2 / r$
 $v = Bq r / m = (0.30)(1.60 \times 10^{-19})(0.10) / (1.67 \times 10^{-27}) = 2.87 \times 10^6 \text{ m s}^{-1}$
 $KE = \frac{1}{2} m v^2 = 6.898 \times 10^{-15} \text{ J} = 43 \text{ keV}$

- 16 B The compass needle points in direction of the resultant B-field of solenoid and that of Earth.
 B-field due to solenoid is proportional to the current passing through it and is in direction East (as the resultant is between North and East)



Initial, since angle is 45° , $B_{\text{solenoid}} = B_{\text{Earth}}$

Final, $\tan \theta = 2 B_{\text{solenoid}} / B_{\text{Earth}} = 2 B_{\text{solenoid}} / B_{\text{solenoid}} = 2$
 $\theta = 63^\circ$

- 17 D The magnetic field is in direction parallel to motion of the electron, hence no magnetic force.
- 18 C Electron is negatively charged, hence the magnetic force is in the direction C (using FLHR). The electric force therefore needs to be in the direction B in order for the electron to have zero resultant force. The direction of E-field will therefore has to be in direction C (electric field direction is opposite to direction of force experienced by a negatively charge placed in that electric field).
- 19 D The magnetic force acting on the wire is to the East (using FLHR). By Newton's 3rd law, the magnet will experience a force to the West.

20 A Test and check, starting with option A: Resultant B-field = $\frac{\mu_0(4.2)}{2\pi(0.15)} - \frac{\mu_0(0.40)}{2\pi(0.05)} = 4.0 \times 10^{-6} \text{ T}$

21 A B-field due to vertical wire : $B = \frac{\mu_0 I}{2\pi d} = \frac{\mu_0(4)}{2\pi(3)} = 2.667 \times 10^{-7}$ out of paper

B-field due to horizontal wire : $B = \frac{\mu_0 I}{2\pi d} = \frac{\mu_0(2)}{2\pi(3)} = 1.333 \times 10^{-7}$ into paper

Resultant B-field is difference between the two = 1.33×10^{-7} out of paper

- 22 C Using moments about the pivots : $(BIL)(15) = mg(2)$

$$\left(\frac{\mu_0 N I_c}{2r} \right) I L (15) = mg (2)$$

$$\left(\frac{4\pi \times 10^{-7} \times 1000 \times I_c}{2 \times 0.125} \right) (5) (0.04)(15) = (0.005)(9.81)(2)$$

$$I_c = 6.5 \text{ A}$$

- 23 B Note that the diagram is a top view. Current has to flow from P to Q such that the magnetic force is out of paper (to balance its weight) using FLHR.

$$mg = BIL \sin 30$$

$$I = mg / BL \sin 30 = (0.50) / (1.0 \times 0.50 \times \sin 30) = 2.0 \text{ A}$$

- 24 C Since it is alternating current, assume a certain direction of the current, and use FLHR to work out the direction of the magnetic force on the straight portion. Even when the current reverses direction, the direction of current in the coil and the straight portion also reverse direction, making the direction of the magnetic force still downwards, in direction Y.

25 C Torque = $F \times d = NBIL \times d = (20)(0.010)(0.005)(0.008)(0.008) = 6.4 \times 10^{-8} \text{ N m}$

26 D $Bqv = mv^2 / r \Rightarrow r = mv / Bq$

When v decreases, r decreases. Hence it spirals inwards (anti-clockwise).

Electron is negatively charged. Using FLHR, the B-field has to be out of paper such that the electron will move as indicated by the diagram (anti-clockwise).

- 27 C Direction can be determined using FLHR.

$$\text{Torque} = F \times d = NBIL \times d = (200)(0.35)(1.6)(0.05)(0.04) = 0.224 \text{ N m}$$

Structured Questions

1. [CJC/H2/Prelims 2010/P3/1]

- a. A field is a region of space in which an object placed in that region, with certain properties, will experience a force associated with the field.

b. i. $+1.60 \times 10^{-19} \text{ C}$

- ii. By conservation of energy,

$$-q\Delta V = \frac{1}{2}mv^2 - 0$$

$$-(1.60 \times 10^{-19}) \times (-1400) = \frac{1}{2} \times [20 \times (1.66 \times 10^{-27})] v^2$$

$$v = 1.16 \times 10^5 \text{ m s}^{-1}$$

- iii. 1. Vertically downward

2. Into the plane of the paper

iv. $F_e = F_b$

$$qE = qBv$$

$$B = \frac{E}{v} = \frac{6.2 \times 10^3}{1.1616 \times 10^5} = 0.0534 \text{ T}$$

- c. i. Greater

- ii. Curving upwards / towards the positive plate (due to $F_b > F_e$).

2. [CJC/H2/Prelims 2010/P3/6 (part)]

- a. Refer to your lecture notes for definitions.

b. i. $N = \frac{200}{0.050} = 4000$

ii. $L = N(\pi d) = 4000 \times \pi \times (0.030) = 376.99 \text{ m}$

$$R = (8.66) \times (376.99) = 3260 \Omega$$

iii. $n = \frac{N}{L} = \frac{4000}{200} = 20.0 \times 10^3 \text{ m}^{-1}$

$$I = \frac{V}{R} = \frac{6.0 \text{ V}}{3264.7 \Omega} = 1.8378 \times 10^{-3} \text{ A}$$

$$B = \mu_0 n I = (4\pi \times 10^{-7}) \times (20000) \times (1.8378 \times 10^{-3}) = 46.2 \times 10^{-6} \text{ T}$$

- c. i. The magnetic flux density increases significantly. The ferromagnetic property of the iron filing causes the internal magnetic domains of the iron to line up with the smaller driving magnetic field produced by the current in the solenoid, thus multiplying the magnetic flux density produced by the solenoid by factors of tens to even thousands.
- ii. The magnetic flux density at the centre of the solenoid is unchanged. With two layers of wire, the number of turns per unit length n is doubled, and the total resistance is also doubled. Thus, the current I flowing in the wire is halved. Hence, B is unchanged.

3. [Dunman High/H2/Prelims 2010/P2/4]

- a. i. Negative

- ii. In the magnetic field, the particle follows a circular path with a uniform speed v_0 .

In the electric field, the particle experiences an electric force parallel to the y -axis and moves in a parabolic path.

b. i. Magnetic force provides the centripetal force. $qv_0B = \frac{mv_0^2}{r} \Rightarrow r = \frac{mv_0}{Bq} \Rightarrow v_0 = \frac{Bqr}{m}$

Time taken for the particle to move in a quadrant of a circle is: $t = \frac{1}{4} \times \frac{2\pi r}{v_0} = \frac{2\pi r}{4} \times \frac{m}{Bqr} = \frac{\pi m}{2Bq}$

ii. $F_e = qE$

$$a_y = \frac{F_e}{m} = \frac{qE}{m}$$

$$s_y = u_y t + \frac{1}{2} a_y t^2$$

$$r = \frac{1}{2} a_y t^2 \text{ (the particle travels a distance of } r \text{ in the negative } y\text{-direction)}$$

$$t = \sqrt{\frac{2r}{a_y}} = \sqrt{\frac{2rm}{qE}}$$

Substitution of $r = \frac{mv_0}{Bq}$ gives $t = \sqrt{\frac{2m^2 v_0}{q^2 B E}} = \frac{m}{q} \sqrt{\frac{2v_0}{B E}}$

- c. i. Work done by the magnetic force is zero since the velocity of the particle is perpendicular to the magnetic force.

- ii. Work done by the electric force is equal to the change in kinetic energy:

$$W = \Delta E_k = \frac{1}{2} m (v^2 - v_0^2)$$

4. [HCL/H2/Prelims 2010/P3/6 (modified)]

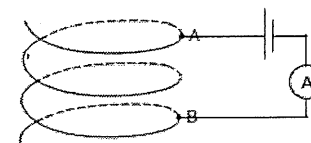
- a. Refer to your lecture notes for definitions.

b. i. $B_1 = \frac{\mu_0 I_1}{2\pi d}, \quad B_2 = \frac{\mu_0 I_2}{2\pi d}$

$$F_{21} = B_1 I_2 L, \quad F_{12} = B_2 I_1 L$$

$$\frac{F}{L} = \frac{F_{21}}{L} = B_1 I_2 = \frac{\mu_0 I_1 I_2}{2\pi d} = B_2 I_1 = \frac{F_{12}}{L}$$

- ii. 1.



2. The current in each turn of the coil of the spring produces a magnetic field that is perpendicular to the current in the adjacent coil.

Since the current in the adjacent spring coils is flowing in the same direction, an attractive force will be exerted on the coils towards each other.

The spring coils will move towards one another and the length (vertical) will shorten. The change in the length of the spring is therefore a compression.

3. Compression

- c. i. According to Faraday's law of Electromagnetic Induction, when the metal axle Y of the truck first enters the magnetic field entry point P with an initial velocity, there will be an induced electromotive force (e.m.f.) across the two metal wheels of Y whose magnitude is directly proportional to the rate of change of magnetic flux linkage. The wheel on the left (at the top according to figure) will be at a higher potential.

Since the axle Y, the two wheels and the railing form a complete loop, induced current will hence flow.

Using Fleming's left hand rule, a magnetic force opposite to the direction of motion would be produced due to this current flow. This opposing force on the wheels will cause braking to occur.

ii. $E = IR = B \times \frac{dA}{dt} = Bwv$

$$I_{\text{induced}} = \frac{E}{R} = \frac{Bwv}{R}$$

Take the direction of motion to be positive,

$$F_B = -BI_{\text{induced}}w \Rightarrow Ma = -B\left(\frac{Bwv}{R}\right)w \Rightarrow a = -\left(\frac{B^2w^2}{MR}\right)v \Rightarrow a \propto v$$

- iii. To improve, we can

EITHER have several regions of PQ placed close to each other for consecutive braking effect.

OR have a stronger vertical magnetic field B and any reasonable explanation that leads to a larger acceleration hence braking force.

(If the truck's initial speed is high, the magnetic braking force experienced upon entering and leaving the magnetic field may not be able to bring the truck to rest completely.)

5. [HCI/H2/Prelims 2010/P3/7]

- a. Similarities: any two

- Both fields exert forces on moving charged particles.
- They are non-contact forces / action-at-a-distance force.
- They are conservative fields.

Differences: any two

- Electric field will change the magnitude of the speed of charged particles whereas magnetic field can be used to change the direction of motion of charged particles.
- If the paths of charged particles are not straight line path, it will be parabolic and circular paths in electric and magnetic field respectively.
- The electric force is either parallel or anti-parallel to the electric field whereas the magnetic force is always mutually perpendicular to the magnetic field and the motion of charged particle.
- There will not be any magnetic force on stationary charged particle whilst there will be electric force on charged particle regardless of its state of motion.

- b. i. Using Fleming's Left Hand Rule, the magnetic force will be mutually perpendicular to the velocity and the magnetic field.

It provides the necessary centripetal force for uniform circular motion.

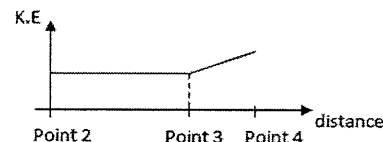
- ii. magnetic force provides the centripetal force

$$Bqv = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{qB}$$

- iii.

$$T = \frac{2\pi r}{v} = \frac{2\pi \left(\frac{mv}{qB}\right)}{v} = \frac{2\pi m}{qB}$$

- iv.



v. 1. $\Delta E_k = 2 \times eV = 2 \times (1.60 \times 10^{-19}) \times 100 = 3.20 \times 10^{-17} \text{ J}$

2. $T = \frac{2\pi m}{qB} \Rightarrow f = \frac{qB}{2\pi m} = \frac{(1.60 \times 10^{-19})(0.400)}{2\pi \times (3.34 \times 10^{-27})} = 3.05 \times 10^6 \text{ Hz}$

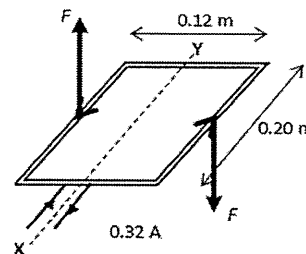
3. Number of revolutions to gain 1.00 MeV = $\frac{1.00 \times 10^6 \times 1.60 \times 10^{-19} \text{ J}}{3.20 \times 10^{-17} \text{ J}} = 5000$

$$t = 5000 \times T = \frac{5000}{f} = \frac{5000}{3.04968 \times 10^6} = 1.64 \times 10^{-3} \text{ s}$$

6. [IJC/H2/Prelims 2010/P2/4]

- a. i. Refer to your lecture notes for definitions.
ii. not in syllabus

- b. i.



ii. $F = NBIL \sin \theta = 150 \times 0.36 \times 0.32 \times 0.20 \times \sin 90^\circ = 3.46 \text{ N}$

iii. $\tau = Fd = 3.456 \times 0.12 = 0.415 \text{ N m}$

7. [IJC/H2/Prelims 2010/P3/3]

- a. The magnitude of the electric force is independent of the speed of the charged particle while the magnitude of the magnetic force is directly proportional to the speed of the charged particle.

The direction of electric force is along the direction of the electric field while the direction of the magnetic force is perpendicular to the direction of the magnetic field.

- b. i. out of the plane of paper
 ii. magnetic force provides centripetal force

$$Bqv = \frac{mv^2}{r}$$

$$r = \frac{mv}{qB} = \frac{(1.67 \times 10^{-27}) \times (4.5 \times 10^6)}{(1.60 \times 10^{-19}) \times (0.12)} = 0.391 \text{ m}$$

- c. i. Arrow pointing vertically upwards

ii. $F_E = F_B$
 $qE = Bqv$

$$E = vB = (4.5 \times 10^6) \times (0.12) = 5.40 \times 10^5 \text{ V m}^{-1}$$

- d. The magnitude of gravitational force is negligible as compared to the magnitude of the electric or magnetic forces. ('Mass of proton is small' is an insufficient answer.)

8. [MI/H2/Prelims 2010/P2/5]

- a. Refer to your lecture notes for definitions.

- b. i. Current flowing in each coil of the spring creates a magnetic field.

The magnetic field of one coil interacts with the magnetic field of adjacent coil resulting in magnetic forces between coils.

- ii. Current in each coil flows in the same direction. The spring is compressed due to attractive magnetic forces between the coils.

- c. i. Into the paper direction indicated with a $\otimes$

- ii. Larger radius of curvature – above the given path.

- d. i. $F_B = Bqv = (1.60 \times 10^{-19}) \times (8.4 \times 10^{-3}) \times (3.4 \times 10^7) = 4.57 \times 10^{-14} \text{ N}$

- ii. Work done on electron by the magnetic force is zero since the magnetic force acting on the electron is perpendicular to its displacement. Hence, the kinetic energy of the electron remains unchanged.

9. [MJC/H2/Prelims 2010/P3/7]

- a. i. Circular path with upward deflection

- ii. The magnetic force acting on the α -particle is directed at right angles to the velocity of the particle.

The resulting acceleration is directed at right angles to the velocity of the particle and thus will not alter the speed but merely changes its direction. **OR** The magnetic force hence does not do work on the particle.

The kinetic energy of the particle thus does not change.

- iii. loss in E.P.E. = gain in K.E.

$$q\Delta V = \frac{1}{2}mv^2 - 0$$

$$2 \times (1.60 \times 10^{-19}) \times (4.0 \times 10^3) = \frac{1}{2} \times (6.644 \times 10^{-27}) \times v^2$$

$$v = 6.21 \times 10^5 \text{ m s}^{-1}$$

iv. $F_B = F_E$

$$Bqv = qE$$

$$E = vB = (6.2073 \times 10^5) \times 2.00 = 1.24 \times 10^6 \text{ V m}^{-1}$$

E field is directed downwards.

- b. i. magnetic force provides the centripetal force

$$Bqv = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{qB} = \frac{(6.644 \times 10^{-27}) \times (6.2073 \times 10^5) \times \sin 30^\circ}{(2 \times 1.60 \times 10^{-19}) \times (2.00)} = 3.22 \times 10^{-3} \text{ m}$$

ii. $Bq_\alpha v = m_\alpha v \omega \Rightarrow Bq_\alpha = m_\alpha \times \frac{2\pi}{T} \Rightarrow T = \frac{2\pi m_\alpha}{q_\alpha B}$

iii. $p = v_{\text{hor}} T = (v \cos 30^\circ) \times \frac{2\pi m_\alpha}{q_\alpha B} = (6.207 \times 10^5) \cos 30^\circ \times \frac{2\pi \times (6.644 \times 10^{-27})}{(2 \times 1.60 \times 10^{-19}) (2.00)} = 0.0351 \text{ m}$

- iv. For a positron, the period $T = \frac{2\pi m}{qB}$ will decrease because

$$\frac{m_\alpha}{q_\alpha} = \frac{6.644 \times 10^{-27}}{2 \times 1.60 \times 10^{-19}} = 2.07 \times 10^{-8} \text{ kg C}^{-1}$$

$$\frac{m_{\text{positron}}}{q_{\text{positron}}} = \frac{9.11 \times 10^{-31}}{1.60 \times 10^{-19}} = 5.69 \times 10^{-12} \text{ kg C}^{-1} \ll 2.07 \times 10^{-8} \text{ kg C}^{-1}$$

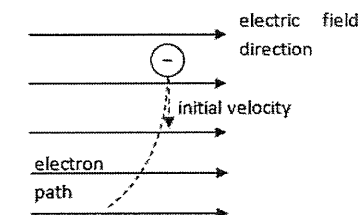
For the same reason, the radius $r = \frac{mv}{qB}$ will also decrease.

Since period decreases, the pitch $p = v_{\text{hor}} T$ will decrease.

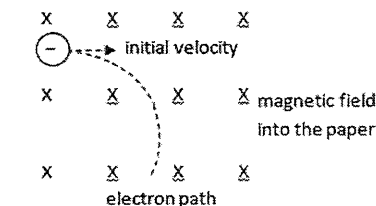
10. [PJC/H2/Prelims 2010/P2/4]

- a. refer to Q1

- b. i. The electrons move in parabolic paths towards the higher potential region.



- ii. The electrons move in circular paths.



- c. i. As current flows through the conductor, the electrons experience a downward magnetic force, according to Fleming's left hand rule. Negative charge accumulates on SR, leaving a net negative charge on SR and a net positive charge on PQ. The voltage is set up across the sides. This voltage eventually stops further vertical migration of electrons.

- ii. Since eventually, electrons are no longer deviated in the magnetic and electric fields,

$$F_E = F_B$$

$$qE = Bqv$$

$$q \frac{\Delta V}{x} = Bqv$$

$$\Delta V = Bvx = (1.0)(2.52 \times 10^{-5})(5 \times 10^{-3}) = 1.26 \times 10^{-7} \text{ V}$$

11. [RI/H2/Prelims 2010/P3/4]

- a. magnetic force provides centripetal force

$$Bqv = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{Bq}$$

$$T = \frac{2\pi r}{v} = \frac{2\pi}{v} \left(\frac{mv}{Bq} \right) = \frac{2\pi m}{Bq}$$

- b. i. $T = \frac{2\pi \times (6.68 \times 10^{-27})}{(0.850) \times (2 \times 1.60 \times 10^{-19})} = 1.54 \times 10^{-7} \text{ s}$

- ii. In order for the nucleus to accelerate when it crosses the gap, frequency of the alternating voltage = orbital frequency of the nucleus.

$$f = \frac{1}{T} = \frac{1}{1.5431 \times 10^{-7}} = 6.48 \times 10^6 \text{ Hz}$$

- iii. K.E. after one revolution = Work done by E-field on helium nucleus = $2qV = 2(2e)V = 4eV$

- iv. The gain in K.E. after each revolution = $4eV$

Hence, the gain in K.E. after five revolutions = $20eV$

$$\frac{1}{2}mv^2 - 0 = 20eV$$

$$v = \sqrt{\frac{2 \times 20 \times (1.60 \times 10^{-19}) \times (450)}{6.68 \times 10^{-27}}} = 6.57 \times 10^5 \text{ m s}^{-1}$$

12. [YJC/H2/Prelims 2010/P2/3]

- a. The constant magnetic force acting on the electron is perpendicular to (the direction of) motion/velocity.

Thus, the magnetic force provides the centripetal acceleration.

And so, the speed of the electron is not changed, but its direction of motion constantly changes, allowing the electron to move in a circle.

- b. magnetic force provides centripetal force

$$Bev = \frac{mv^2}{r} \Rightarrow v = \frac{Ber}{m}$$

$$c. \quad v = \frac{Ber}{m} \Rightarrow B = \frac{vm}{er} = \frac{(3.2 \times 10^7) \times (9.11 \times 10^{-31})}{(1.60 \times 10^{-19}) \times (25 \times 10^{-3})} = 7.29 \times 10^{-3} \text{ T}$$

13. [SAJC/H2/Prelims 2010/P3/7]

a. i. $t = \frac{s_{\text{hor}}}{v_{\text{hor}}} = \frac{80.0 \times 10^{-3}}{2.5 \times 10^7} = 3.2 \times 10^{-9} \text{ s}$

ii. $\tan 30^\circ = \frac{v_{\text{ver}}}{v_{\text{hor}}} \Rightarrow v_{\text{ver}} = v_{\text{hor}} \tan 30^\circ = (2.5 \times 10^7) \tan 30^\circ = 1.44 \times 10^7 \text{ m s}^{-1}$

iii. $a = \frac{\Delta v_{\text{ver}}}{\Delta t} = \frac{1.4434 \times 10^7 - 0}{3.2 \times 10^{-9}} = 4.51 \times 10^{15} \text{ m s}^{-2}$

iv. $E = \frac{\Delta V}{d} \Rightarrow \frac{F}{q} = \frac{\Delta V}{d} \Rightarrow \Delta V = d \times \frac{ma}{q} = (40 \times 10^{-3}) \times \frac{(9.11 \times 10^{-31}) \times (4.5105 \times 10^{15})}{1.60 \times 10^{-19}} = 1.03 \times 10^3 \text{ V}$

- b. i. Into the plane of the paper

ii. $F_B = F_E$

$$Bqv = qE$$

$$B = \frac{E}{v} = \frac{\Delta V}{vd} = \frac{1.027 \times 10^3}{(2.5 \times 10^7) \times (40 \times 10^{-3})} = 1.03 \times 10^{-3} \text{ T}$$

- c. i. arc of a circle



- ii. magnetic force provides centripetal force

$$qBv = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{qB} = \frac{(9.11 \times 10^{-31}) \times (2.5 \times 10^7)}{(1.60 \times 10^{-19}) \times (1.027 \times 10^{-3})} = 0.139 \text{ m}$$

- iii. Same speed

The magnetic force, being always perpendicular to the electron's velocity, does not do any work on the electron.

- d. i. The electrons behave like waves.

The electron waves, with wavelengths of the same orders of magnitude as the spacing in the microgrid, diffract upon passing through the microgrid.

The diffracted waves interfere with one another to produce the pattern.

- ii. Using electrons with longer wavelengths / Using lower potentials.

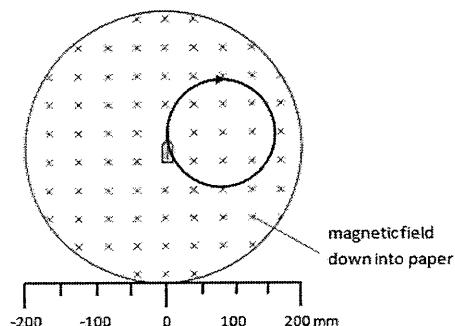
14. [TJC/H2/Prelims 2010/P2/3]

a. Refer to your lecture notes for definitions.

b. i. magnetic force provides centripetal force

$$Bqv = \frac{mv^2}{r} \Rightarrow v = \frac{Bqr}{m} = \frac{(6.5 \times 10^{-4})(1.60 \times 10^{-19})(0.083)}{9.11 \times 10^{-31}} = 9.48 \times 10^6 \text{ m s}^{-1}$$

ii. Circle of diameter roughly 166 mm



iii. The component of velocity along the field causes zero magnetic force so the electron moves at constant velocity along the field.

The component of velocity perpendicular to the field caused a centripetal force so the electron moves in a circular direction.

15. [VJC/H2/Prelims 2010/P3/5]

a. i. Refer to your lecture notes for definitions. SI unit: Tesla (T)

$$ii. B = \frac{F}{IL}$$

$$\text{unit of } B = \frac{\text{N}}{\text{A m}} = \frac{\text{kg m s}^{-2}}{\text{A m}} = \text{kg s}^{-2} \text{ A}^{-1}$$

b. i. Gain in K.E. = Loss in E.P.E.

$$\frac{1}{2}mv^2 = qV$$

$$v = \sqrt{\frac{2qV}{m}}$$

ii. The ion will experience a magnetic force that acts perpendicularly to its velocity.

This force will cause the velocity to change its direction but not its speed.

Consequently, the ion moves in a circular path.

iii. Out of the plane of the paper

iv. magnetic force provides centripetal force

$$Bqv = \frac{mv^2}{r} \Rightarrow r = \frac{mv}{Bq}$$

$$v. x = 2r = \frac{2mv}{Bq} = \frac{2m}{Bq} \sqrt{\frac{2qV}{m}} = \sqrt{\frac{8mV}{B^2q}}$$

c. i. Since ^{14}C is more massive than ^{12}C , then from the equation in b.v. ^{14}C would have a larger diameter in path than ^{12}C . Thus the two isotopes would hit the photographic plate at two spots.

$$ii. \text{ For } ^{14}\text{C}, x_1 = \sqrt{\frac{8 \times (14 \times 1.66 \times 10^{-27}) \times (4000)}{(0.50)^2 \times (1.60 \times 10^{-19})}} = 0.13635 \text{ m}$$

$$\text{ For } ^{12}\text{C}, x_2 = \sqrt{\frac{8 \times (12 \times 1.66 \times 10^{-27}) \times (4000)}{(0.50)^2 \times (1.60 \times 10^{-19})}} = 0.12624 \text{ m}$$

$$\Delta x = x_1 - x_2 = 0.13635 \text{ m} - 0.12624 \text{ m} = 0.0101 \text{ m}$$

d. If an electron were to be introduced, it would be deflected to the left because the electron is negatively charged. The diameter of its path would also be much smaller since the mass of electron is much smaller (only $9.11 \times 10^{-31} \text{ kg}$) (from the equation in b.v.)

III. Solutions and Answers

The suggested solutions and answers below are only for your reference and have not been verified to be error-free. If you have any doubt, please discuss with your tutor to clarify.

MCQs

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
C	A	B	A	D	A	D	A	B	B	B	C	C	D	D

16	17	18	19	20	21	22	23	24	25	26	27	28	29	30
D	D	D	B	D	B	D	A	A	D	B	A	C	A	B

31	32	33
B	C	B

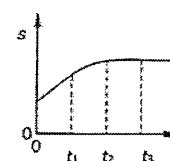
- 1 C Using Fleming's Right Hand Rule, current induced in the metal rod should be in the direction XY. Since XY acts as an emf source, Y should be of a higher potential than X.
Since the metal rod is not connected to any external circuit, it is an open circuit and hence no current flows through it. Thus there should be no magnetic force acting on the rod.
- 2 A The flow of eddy currents in the copper plate oppose the motion of the plate, causing it to come to rest shortly.
Having the slits in the plate reduce the flow of eddy currents because the eddy currents cannot flow across the air gaps in the slits. There is less opposition to the motion of the plate and it takes a longer time to come to rest.
- 3 B $E = -\frac{d\Phi}{dt}$
E.m.f. induced is the negative value of the gradient of the magnetic flux-time graph.
- 4 A Using Fleming's Right Hand Rule, the current induced would flow either from the VWZ plane to the UXY plane or vice versa, depending on the direction of the magnetic field (whether upwards or downwards).
Points V, W and Z will be at the same potential, while U, X and Y will be at the same potential.

5 D $|E| = \left| \frac{d\Phi}{dt} \right| \propto \left| \frac{dB}{dt} \right|$ since $\Phi = NBA \cos \theta$

Induced e.m.f. would be a maximum at the points where the rate of change of flux density B is the highest, i.e. where magnitude of the gradient of the graph is a maximum. (0 ms, 4 ms, 9 ms etc)

- 6 A The same e.m.f. is induced across the ends of the copper bar and across the ends of the voltmeter since they are moving across the magnetic field with the same uniform speed. Hence there is no net emf registered by the voltmeter.

- 7 D The gradient of the $s-t$ graph gives the speed v of the aluminium rod. E.m.f. induced is $E = BLv$



$0 - t_1$: constant speed, constant E

$t_1 - t_2$: decreasing speed, decreasing E

$t_2 - t_3$: speed is zero, $E = 0$

- 8 A The current induced in the solenoid as the bar magnet enters and leaves the coil should be in opposite directions. When the magnet is completely within the solenoid, there is no change in flux linkage (and hence zero induced current) until the magnet starts to exit the solenoid.
- 9 B **Statement 1:** When coil 1 is moved upwards, the flux linkage decreases as the magnetic flux density due to coil 2 is smaller as distance increases. Emf and current is induced in coil 2.
Statement 2: No change in the flux linkage through coil 1. No emf is induced.
Statement 3: When R is increased and decreased rapidly, the flux density of coil 2 decreases and increases respectively. Thus there is a change in flux linkage through coil 1. Emf and current is induced in coil 2.
- 10 B When switch K is closed, a magnetic field is set up in the electromagnet. The ring cuts the magnetic flux of the electromagnet as it swings and a current would be induced in the ring. According to Lenz's Law, the direction of this induced current would oppose the change causing it, and hence oppose the oscillating motion of the ring.
- 11 B Using data from the graph,
 $F = BIL$
 $B = \left(\frac{F}{I} \right) \frac{1}{L} = \left(\frac{10 \times 10^{-3}}{4} \right) \left(\frac{1}{0.50} \right) = 0.004 \text{ T}$
Maximum e.m.f. will be induced when the wire has maximum speed.
Maximum speed of the wire, $v_{\max} = \omega X_0$
 $E_{\max} = BLv_{\max} = (0.004)(0.50)(2\pi 100)(1.5 \times 10^{-3}) = 1.88 \times 10^{-3} \text{ V}$
- 12 C Refer to solution for Q5